AGRICONNECT TN

A direct, price-transparent marketplace connecting Tamil Nadu's farmers to buyers — designed for smartphone users and non-users alike

• Spring Boot backend • Built for real Tamil Nadu farmers

THE PROBLEM

Farmers sell their harvest blind

A real example

A mango farmer near Dindigul harvests his crop each summer. He hires a tempo, drives to the mandi, and hands the produce to a commission agent — with no idea what price other mandis are offering that day.

₹10/kg paid to farmer

₹40/kg paid by the end consumer

Because he arrives with a perishable truckload and no pre-arranged buyer, he has almost no bargaining power once he's at the mandi.

6–10% typical commission agent cut, before market fees

~23% of India's mandis are even connected to e-NAM

#1 ranked farmer complaint: low price at harvest time

Sources: government open agricultural-marketing data and Tamil Nadu farmer-perception studies, 2022–2026

EXISTING SYSTEMS

What's already been tried — and where it falls short

e

e-NAM

Government of India, national trading portal

KEY GAP

Only ~23% of mandis connected; interstate trade volume fell 78% in FY25. Farmers cite the need for immediate cash and distrust of delayed digital payment.

U

Uzhavan app

Tamil Nadu government, 10+ lakh downloads

KEY GAP

Mainly an information/subsidy app, not a live price-negotiation marketplace. Users skew toward already digitally comfortable farmers.

P

Private aggregators

Ninjacart, DeHaat, WayCool & similar

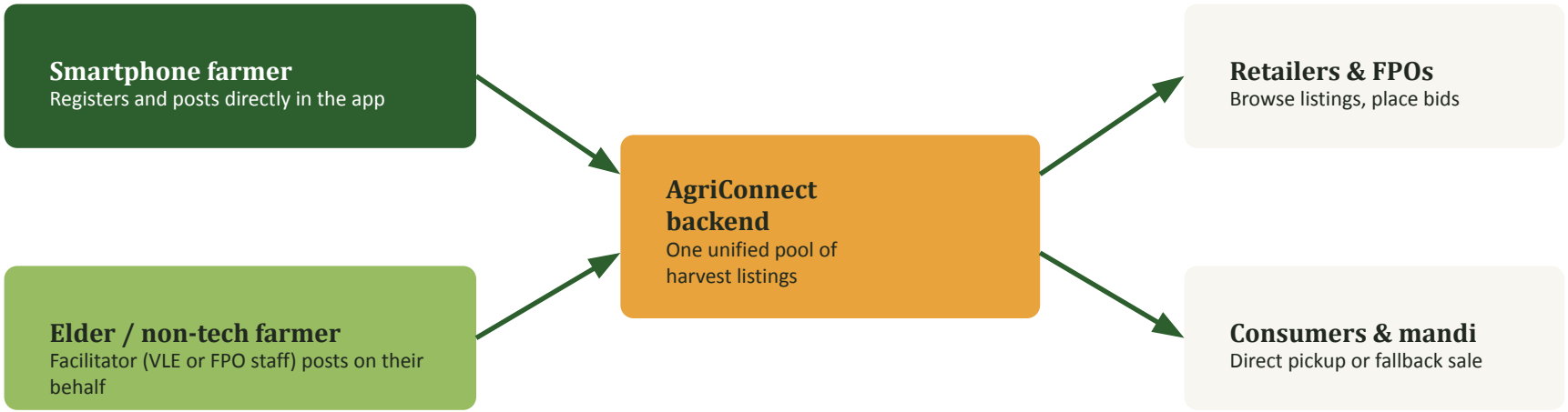
KEY GAP

Re-centralize produce through their own collection agents — a new corporate middleman instead of the traditional one. Often need bulk volumes small farmers can't meet.

OUR APPROACH

One system, two ways in

The farmer is never forced to be the one using a smartphone. Both paths write to the same backend, so price data and buyer demand stay centralized.



OUR APPROACH

Post before harvest, not after

Every existing system reacts to produce that already exists. AgriConnect lets a farmer post an upcoming harvest days or weeks ahead, so buyers can plan instead of the farmer arriving at a mandi blind.

1

Draft

Photos + expected window, not visible yet

2

Posted

Buyers & transporters can browse it

3

Interest received

Multiple offers come in

4

Confirmed

Farmer picks the best match

5

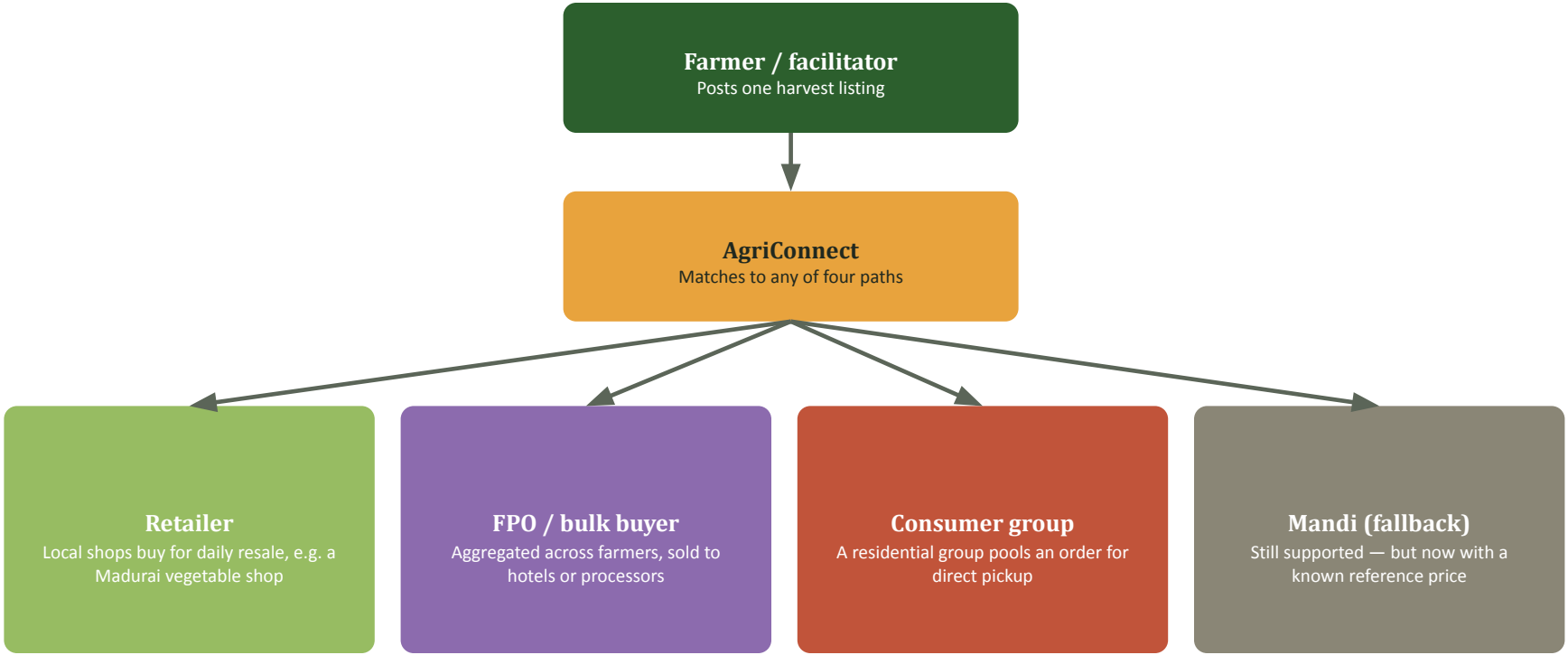
Fulfilled

Sale price and rating recorded

↻ *Every fulfilled listing records a real sale price — building the platform's own price-transparency data over time, without depending on mandi price boards.*

OUR APPROACH

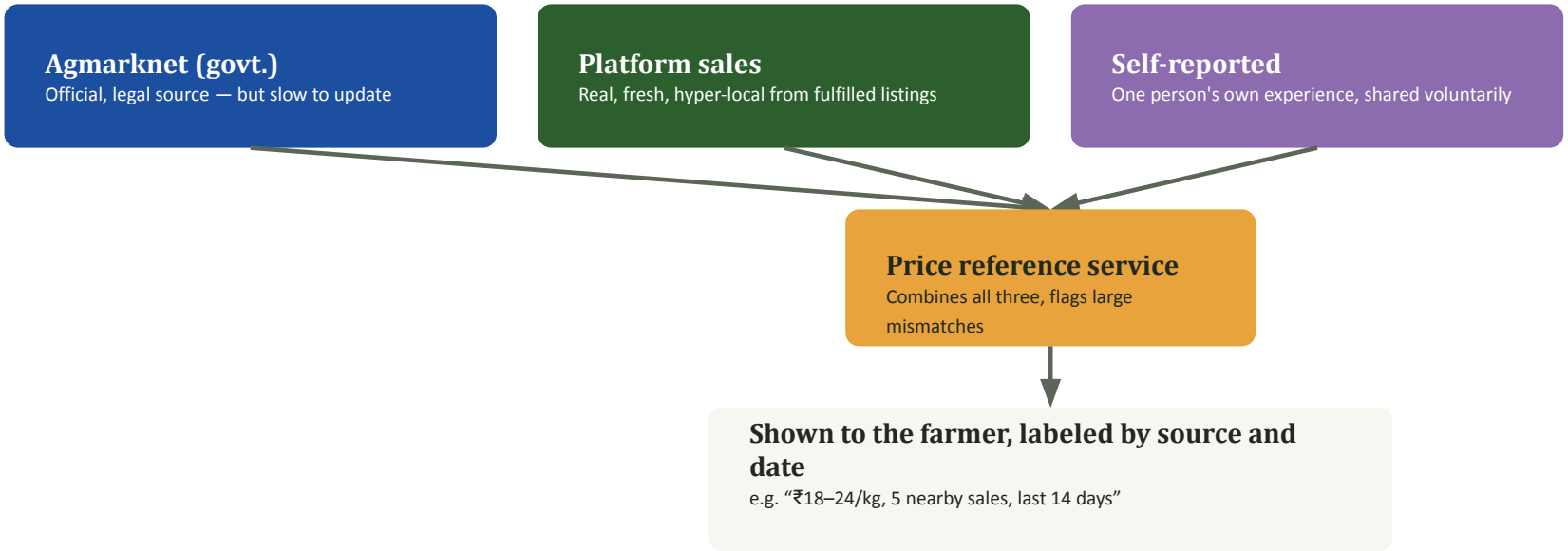
One listing, four buyer paths



OUR APPROACH

Price data from three sources, never just one

Mandi price boards are often stale or under-reported, and asking anyone to photograph an agent's board risks real friction. So no single source is trusted blindly.



HOW IT'S BUILT

Spring Boot backend, entirely on free-tier infrastructure

Backend

Spring Boot 3, Spring Security (JWT), Spring Data JPA

Hosting

Azure App Service, F1 free tier

Database

PostgreSQL — Azure free tier / Neon.tech

CI/CD

GitHub Actions, auto-deploy on push

Cache

Redis — Upstash free tier

Docs & monitoring

OpenAPI/Swagger, Application Insights

Core entities: User (role-based) · HarvestListing · Interest · PickupSchedule · Outcome · PriceReference

BUILD PLAN

Eight-phase build roadmap

01

Accounts & tools setup

02

Scaffold Spring Boot project

03

Auth + Farmer/Buyer module

04

Harvest listing + search

05

Connect free cloud database

06

Orders, FPO, price reference

07

CI/CD and deployment

08

Polish & demo readiness

Why this matters

More of every harvest's value stays with the farmer who grew it.

Built as a hands-on demonstration of designing for real constraints — low digital literacy, unreliable price data, and the trust dynamics of rural India — not just a CRUD app with a farming theme.

Thank you